Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Andrew Zylstra Dairy 2

Physical address of dairy:

7855 S Carpenter RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/01/1994

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X002-X016-XXXX

B. OPERATORS

Zylstra, Michael John

Operator name: Zylstra, Michael John

Telephone no.: (209) 535-2883 (Landline) (Cellular)

2655 W Monte Vista AVE	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Zylstra, Michael John

Legal owner name: Zylstra, Michael John

Telephone no.: (209) 535-2883 (Landline) (Cellular)

2655 W Monte Vista AVE	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	102	0	0	0
Number under roof	650	0	0	0	0	0
Maximum number	690	0	102	0	0	0
Average number	650	0	102	0	0	0
Avg live weight (lbs)	1,400	0	900	0		

Predominant milk cow breed: Holstein

Average milk production: 86 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,934.24 *tons per reporting period*

Total nitrogen from manure: 243,635.25 *lbs per reporting period*

After ammonia losses (30% loss applied): 170,544.68 *lbs per reporting period*

Total phosphorus from manure: 41,478.02 *lbs per reporting period*

Total potassium from manure: 131,313.80 *lbs per reporting period*

Total salt from manure: 306,052.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,729,600 *gallons*

Total nitrogen generated: 51,202.14 *lbs*

Total phosphorus generated: 3,781.74 *lbs*

Total potassium generated: 63,312.19 *lbs*

Total salt generated: 786,090.99 *lbs*

	12,729,600	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,729,600	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
TID Lateral 3	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Dairy 2	61	61	2	both	0022-0002-0016-0000
Simmons	50	50	2	process wastewater	X022-X002-X010-XXXX
Totals for areas that were used for application	111	111	4		
Totals for areas that were not used for application					
Land application area totals	111	111	4		

B. CROPS AND HARVESTS

Dairy 2

Field name: Dairy 2

11/04/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 61 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	952.00 *ton*	As-is		70.0	8,820.00	1,120.00	10,900.00		10.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	15.61	275.30	34.96	340.22	964.49

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 61 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	1,767.00 *ton*	As-is		67.6	3,950.00	620.00	3,850.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.97	228.84	35.92	223.05	882.23

Simmons

Simmons

Field name: Simmons

11/02/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 50 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	899.00 *ton*	As-is		70.0	4,880.00	1,390.00	10,500.00		8.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	17.98	175.48	49.98	377.58	938.56

05/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 05/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,565.00 *ton*	As-is		66.2	3,450.00	530.00	5,000.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.30	215.97	33.18	313.00	1,100.26

NUTRIENT BUDGET

A. LAND APPLICATIONS

Dairy 2 - 11/04/2022: Triticale, soft dough

Field name: Dairy 2

Crop: Triticale, soft dough

Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.22	3.86	64.58	801.78	792,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	24.36	6,596,370.00 *gal*
Application event totals		52.22	3.86	64.58	826.14	

10/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	144.05	63.22	83.58	6,393.21	600.00 *ton*
Application event totals		144.05	63.22	83.58	6,393.21	

02/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.96	5.61	93.93	1,166.22	1,152,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	15.96	4,320,000.00 *gal*
Application event totals		75.96	5.61	93.93	1,182.18	

03/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	56.97	4.21	70.45	874.67	864,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.58	7,196,400.00 *gal*
Application event totals		56.97	4.21	70.45	901.25	

Dairy 2 - 05/25/2023: Corn, silage

Dairy 2 - 05/25/2023: Corn, silage

Field name: Dairy 2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Manure	Separator solids	90.03	39.51	52.29	3,995.75	375.00 *ton*
Application event totals		90.03	39.51	52.29	3,995.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.94	5.09	85.24	1,058.35	1,045,440.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	34.62	9,373,676.00 *gal*
Application event totals		68.94	5.09	85.24	1,092.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	64.38	17,430,390.00 *gal*
Application event totals		0.00	0.00	0.00	64.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.11	4.37	73.09	907.47	896,400.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.69	8,037,347.00 *gal*
Application event totals		59.11	4.37	73.09	937.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	44.95	12,168,995.00 *gal*
Application event totals		0.00	0.00	0.00	44.95	

Dairy 2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.65	3.82	63.87	793.03	783,360.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.94	7,023,802.00 *gal*
Application event totals		51.65	3.82	63.87	818.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.99	3.84	64.28	798.13	788,400.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.11	7,068,992.00 *gal*
Application event totals		51.99	3.84	64.28	824.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	25.60	6,931,872.00 *gal*
Application event totals		0.00	0.00	0.00	25.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	6.37	1,723,672.00 *gal*
Application event totals		0.00	0.00	0.00	6.37	

Simmons - 11/02/2022: Triticale, soft dough

Field name: Simmons

Crop: Triticale, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Simmons - 11/02/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/08/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.09	5.99	100.27	1,244.94	1,008,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	36.38	8,074,080.00 *gal*
Application event totals		81.09	5.99	100.27	1,281.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.51	5.13	85.94	1,067.09	864,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	31.19	6,920,640.00 *gal*
Application event totals		69.51	5.13	85.94	1,098.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.51	5.13	85.94	1,067.09	864,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	31.19	6,920,640.00 *gal*
Application event totals		69.51	5.13	85.94	1,098.28	

Simmons - 05/07/2023: Corn, silage

Field name: Simmons

Crop: Corn, silage

Plant date: 05/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.67	6.84	114.59	1,422.79	1,152,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	41.58	9,227,520.00 *gal*
Application event totals		92.67	6.84	114.59	1,464.37	

Simmons - 05/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	49.38	10,957,680.00 *gal*
Application event totals		0.00	0.00	0.00	49.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	35.94	7,976,038.00 *gal*
Application event totals		0.00	0.00	0.00	35.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.71	4.71	78.78	978.17	792,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.59	6,343,920.00 *gal*
Application event totals		63.71	4.71	78.78	1,006.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.30	5.56	93.11	1,156.02	936,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	33.79	7,497,360.00 *gal*
Application event totals		75.30	5.56	93.11	1,189.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.19	6,920,640.00 *gal*
Application event totals		0.00	0.00	0.00	31.19	

Simmons - 05/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.71	4.71	78.78	978.17	792,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.59	6,343,920.00 *gal*
Application event totals		63.71	4.71	78.78	1,006.76	

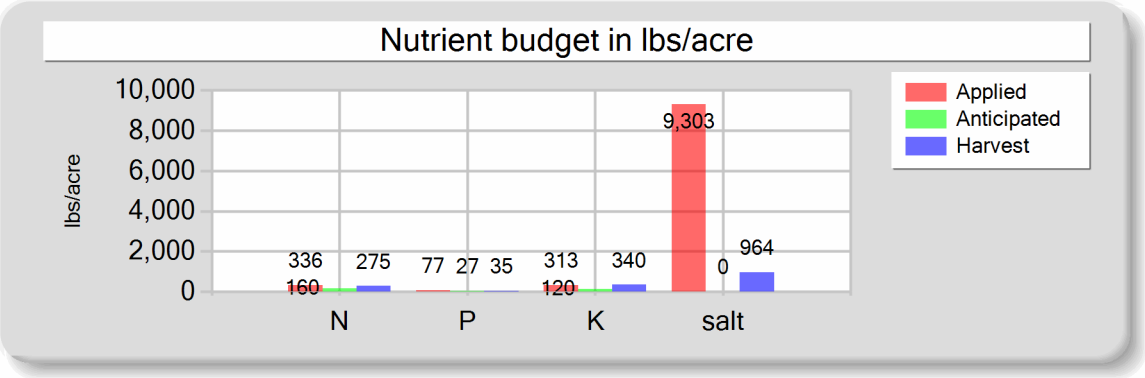
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.98	8,650,800.00 *gal*
Application event totals		0.00	0.00	0.00	38.98	

B. NUTRIENT BUDGET

Dairy 2 - 11/04/2022: Triticale, soft dough

Field name: Dairy 2 Crop: Triticale, soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	144.05	63.22	83.58	6,393.21
Process wastewater	185.16	13.68	228.95	2,842.66
Fresh water	0.00	0.00	0.00	66.90
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	336.20	76.90	312.53	9,302.77
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	275.30	34.96	340.22	964.49
Nutrient balance	60.90	41.94	-27.69	8,338.29
Applied to removed ratio	1.22	2.20	0.92	9.65

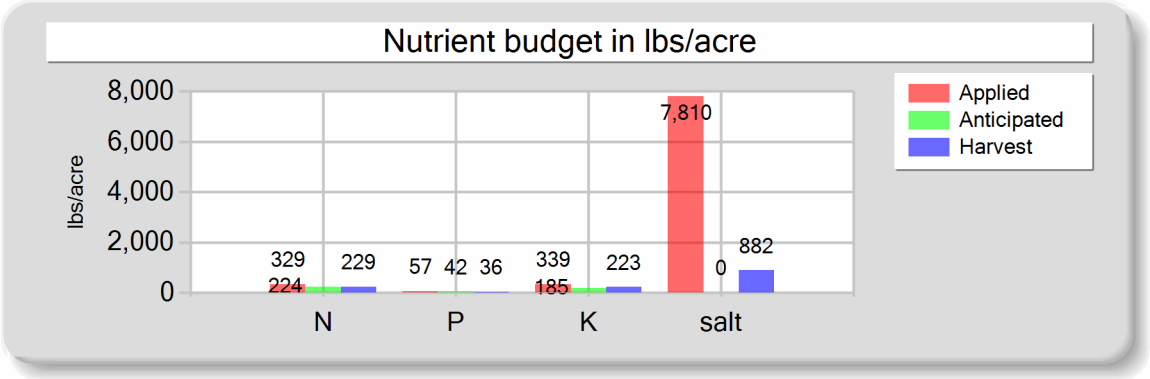
Fresh water applied	
18,112,770.00	*gallons*
667.03	*acre-inches*
10.93	*inches/acre*

Process wastewater applied	
2,808,000.00	*gallons*
103.41	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 2 - 05/25/2023: Corn, silage

Field name: Dairy 2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	90.03	39.51	52.29	3,995.75
Process wastewater	231.68	17.11	286.48	3,556.97
Fresh water	0.00	0.00	0.00	257.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	328.71	56.62	338.78	7,810.39
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	228.84	35.92	223.05	882.23
Nutrient balance	99.87	20.71	115.73	6,928.17
Applied to removed ratio	1.44	1.58	1.52	8.85

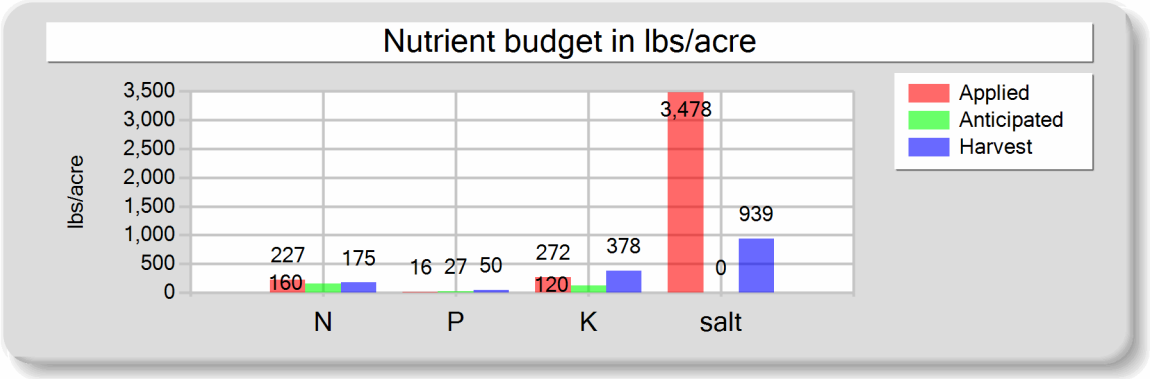
Fresh water applied	
69,758,746.00	*gallons*
2,568.98	*acre-inches*
42.11	*inches/acre*

Process wastewater applied	
3,513,600.00	*gallons*
129.39	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

Simmons - 11/02/2022: Triticale, soft dough

Field name: Simmons | Crop: Triticale, soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.10	16.26	272.16	3,379.12
Fresh water	0.00	0.00	0.00	98.76
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.10	16.26	272.16	3,477.88
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	175.48	49.98	377.58	938.56
Nutrient balance	51.61	-33.73	-105.42	2,539.33
Applied to removed ratio	1.29	0.33	0.72	3.71

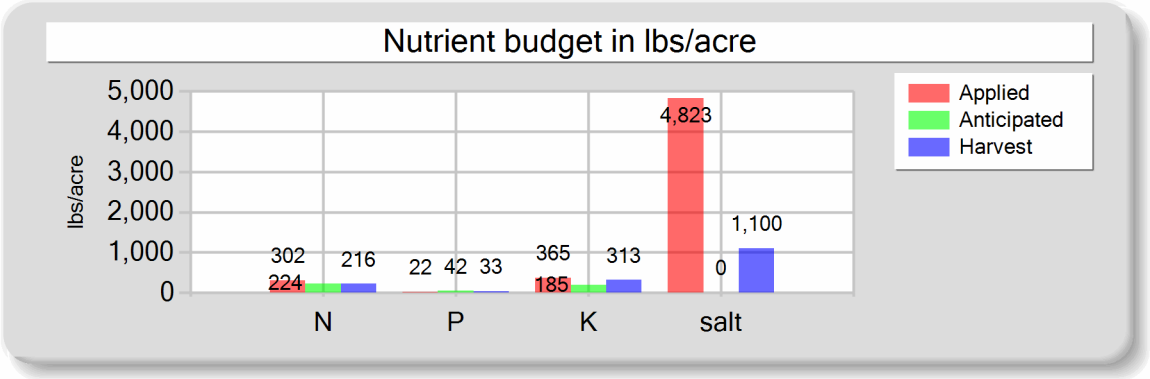
Fresh water applied	
21,915,360.00	*gallons*
807.07	*acre-inches*
16.14	*inches/acre*

Process wastewater applied	
2,736,000.00	*gallons*
100.76	*acre-inches*
2.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

Simmons - 05/07/2023: Corn, silage

Field name: Simmons Crop: Corn, silage Plant date: 05/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	295.40	21.82	365.26	4,535.14
Fresh water	0.00	0.00	0.00	288.03
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.40	21.82	365.26	4,823.17
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	215.97	33.18	313.00	1,100.26
Nutrient balance	86.43	-11.36	52.26	3,722.92
Applied to removed ratio	1.40	0.66	1.17	4.38

Fresh water applied	
63,917,878.00	*gallons*
2,353.88	*acre-inches*
47.08	*inches/acre*

Process wastewater applied	
3,672,000.00	*gallons*
135.23	*acre-inches*
2.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	7,110.00	9,410.00	22,200.00	11,100.00	2,210.00	3,920.00	1,400.00		71.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ2

Sample and source description: WWQ2

Sample date: 05/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	893.00	484.00	0.00	0.00	71.70	874.00	280.00	190.00	346.00	4,360.00	0.00	437.00	423.00	8,620.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	430.00	295.00	0.00	0.00	19.70	342.00								6,850.00	4,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	409.00	34.40	0.00	0.00	34.40	504.00								5,380.00	5,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	314.00	0.00	0.00	35.60	596.00								6,210.00	7,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	375.00	246.00	0.00	0.00	35.40	491.00								4,970.00	4,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

TID Lateral 3

TID Lateral 3

TID Lateral 3

Sample description: TID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								35.90	27
DL	1.00	0.50	0.50								10.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Dairy 2 - 11/04/2022: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,820.00	1,120.00	10,900.00		10.30
DL	100.00	100.00	100.00		0.01

Dairy 2 - 05/25/2023: Corn, silage

Dairy 2 - 05/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,950.00	620.00	3,850.00		4.70
DL	100.00	100.00	100.00		0.01

Simmons - 11/02/2022: Triticale, soft dough

Simmons Triticale

Sample and source description: Simmons Triticale

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,880.00	1,390.00	10,500.00		8.70
DL	1.00	1.00	1.00		1.00

Simmons - 05/07/2023: Corn, silage

Simmons Corn

Sample and source description: Simmons Corn

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	530.00	5,000.00		5.20
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

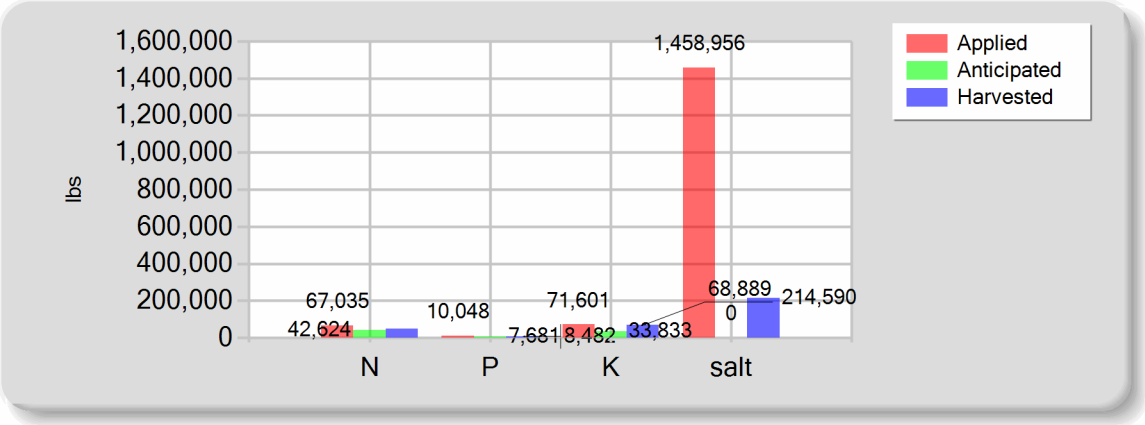
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

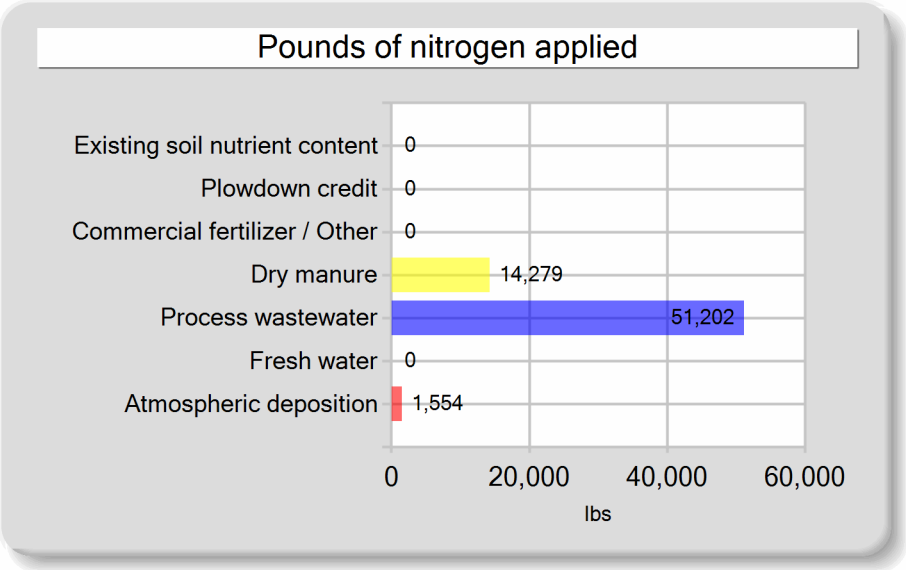
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

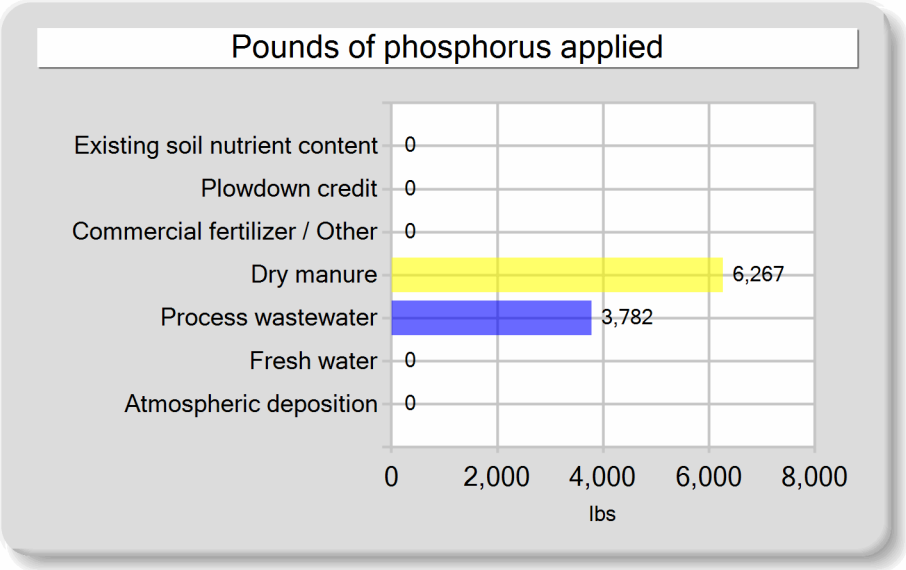
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	14,278.68	6,266.75	8,288.55	633,726.60
Process wastewater	51,202.14	3,781.74	63,312.19	786,090.99
Fresh water	0.00	0.00	0.00	39,138.29
Atmospheric deposition	1,554.00	0.00	0.00	0.00
Total nutrients applied	67,034.82	10,048.49	71,600.74	1,458,955.88
Anticipated crop nutrient removal	42,624.00	7,681.20	33,832.80	0.00
Actual crop nutrient removal	50,325.32	8,481.68	68,888.50	214,590.03
Nutrient balance	16,709.50	1,566.81	2,712.24	1,244,365.84
Applied to removed ratio	1.33	1.18	1.04	6.80

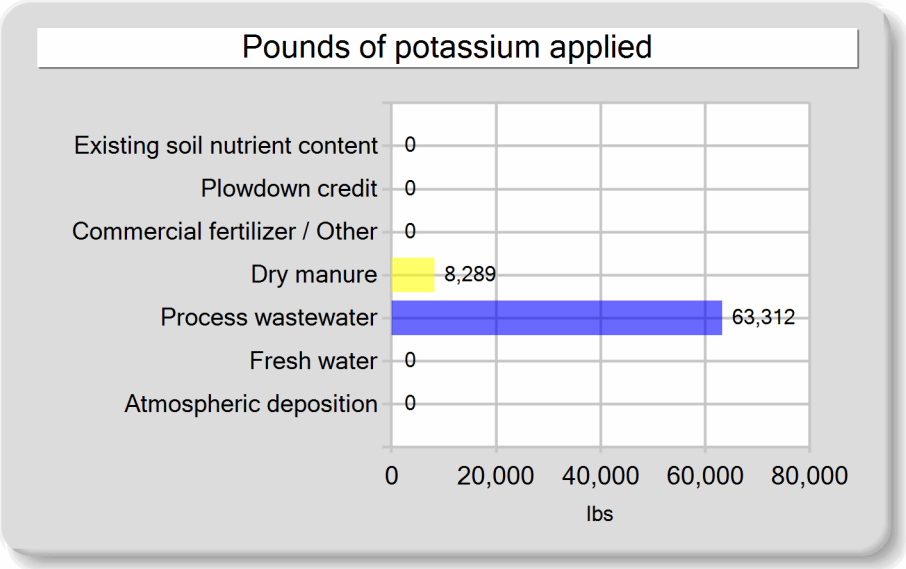
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

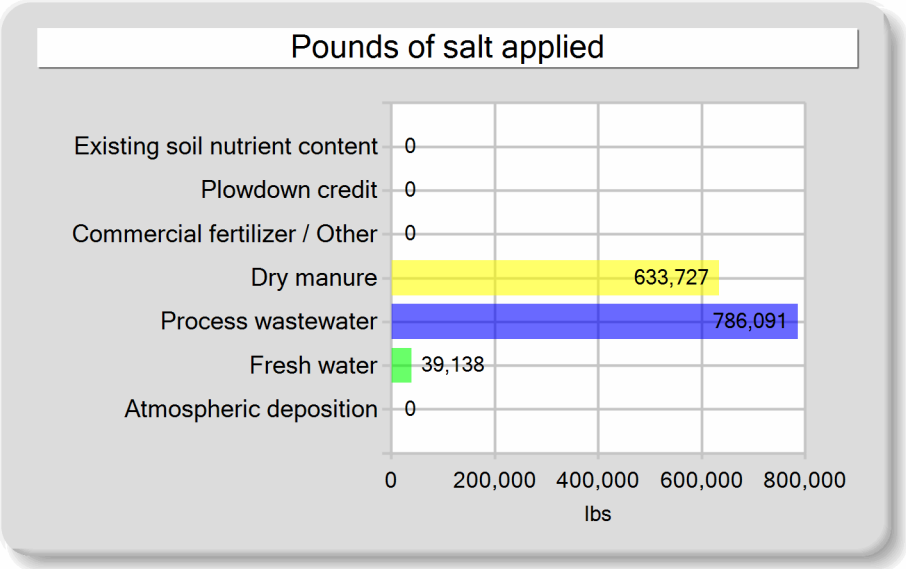


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Mike Zylstra
Mike Zylstra (Jun 26, 2024 09:40 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael John Zylstra	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W6-1 Simmons	T3346003A	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	57.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	821	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W7-1 3501 W. Monte Vista	T3346003B	Electrical Conductivity (EC)	EPA 120.1 and	1.44	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	58.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	922	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W8-1 6807 Jennings (House)	T3346003C	Electrical Conductivity (EC)	EPA 120.1 and	2.60	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	77.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1670	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W8-2 6807 Jennings (Mobile)	T3346003D	Electrical Conductivity (EC)	EPA 120.1 and	2.62	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	76.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1680	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: _____

Client: ANDREW ZYLSTRA DAIRY INC
Mailing address: 2655 W. Monte Vista Ave
(not facility address)
Phone#:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

H1	H3	M3	F18	OTHER

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis				
1	12-12	10:53	W6-1 - Simmons		17.6		H1				
2	↓	↓	W7-1 - 3501 W Monte Vista		17.6		↓				
3			W8-1 - 6807 Jennings (House)		18.8						
4			W8-2 - 6807 Jennings (Mobile)		18.2						
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st	[signature]	Andrew Zylstra	[illegible]			
2nd	[signature]	DENELE ANALYTICAL, INC.	12-12-23 @ 11:59			T3346003

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W1-1 - Dairy 1 - Milk Barn	T3346002A	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	836	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W1-3 - 7800 Vivian	T3346002B	Electrical Conductivity (EC)	EPA 120.1 and	1.96	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1250	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-1 - Dairy 2-A	T3346002C	Electrical Conductivity (EC)	EPA 120.1 and	1.15	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	27.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	737	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-2 - Dairy 2-B	T3346002D	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	707	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-4 - 7855 S. Carpenter	T3346002E	Electrical Conductivity (EC)	EPA 120.1 and	1.60	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	51.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1020	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W3-1 - 2936 W. Monte Vista	T3346002F	Electrical Conductivity (EC)	EPA 120.1 and	1.35	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.09	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	865	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W3-2 - 2848 W. Monte Vista	T3346002G	Electrical Conductivity (EC)	EPA 120.1 and	1.93	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	35.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1240	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 19.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W4-1 - 6618 Jennings	T3346002H	Electrical Conductivity (EC)	EPA 120.1 and	1.56	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	30.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	998	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 20.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W4-2 - 6800 Jennings	T3346002I	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	842	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.6 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W5-1 - 6619 Jennings	T3346002J	Electrical Conductivity (EC)	EPA 120.1 and	2.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	43.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1330	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: _____

Client: ANDREW Zylstra DAIRY INC
Mailing address: 2655 W Monte Vista Ave
(not facility address)
Phone#:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12-12	10:53	W1-1 - DAIRY 1 - MILK BARN		17.8		H1				
2			W1-3 - 7800 Vivian		18.4						
3			W2-1 - DAIRY 2 - A		18.4						
4			W2-2 - DAIRY 2 - B		18.4						
5			W2-4 - 7855 S. CARPENTER		18.4						
6			W3-1 - 2936 W Monte Vista		18.2						
7			W3-2 - 2948 W Monte Vista		17.4						
8			W4-1 - 6618 Jennings		19.8						
9			W4-2 - 6866 Jennings		20.2						
10			W5-1 6619 Jennings		17.6						

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]	Andrew Zylstra Dairy				
2nd		DENELE ANALYTICAL, INC.	12-12-23 @ 11:54			T 3346002

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-1 3831 N Dakota DW	T3353008A	Electrical Conductivity (EC)	EPA 120.1 and	0.837	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	536	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-2 3831 N Dakota DW	T3353008B	Electrical Conductivity (EC)	EPA 120.1 and	0.961	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	615	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

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Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-5 4154 Murphy DW	T3353008C	Electrical Conductivity (EC)	EPA 120.1 and	0.574	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	367	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-4I Dakota IW	T3353008D	Electrical Conductivity (EC)	EPA 120.1 and	0.918	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	555	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	21.1	mg/L
		Soluble Salts (SALT-SOL)	Calculation	587	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)			1.21
Electrical Conductivity (EC)	98.9			1.07 (T3352006A)	1.06 (Dup)	0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)			5.20
Total Dissolved Solids (TDS)	98.5			615 (T3353011A)	620 (Dup)	0.810
Total Nitrogen (TN)	99.7			11.2 (T3338006E)	11.2 (null)	0.000

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Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W9-1I McCarthy IW	T3353008E	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.1	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	545	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	25.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	687	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)			1.21
Electrical Conductivity (EC)	98.9			1.07 (T3352006A)	1.06 (Dup)	0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)			5.20
Total Dissolved Solids (TDS)	98.5			615 (T3353011A)	620 (Dup)	0.810
Total Nitrogen (TN)	99.7			11.2 (T3338006E)	11.2 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: ANDREW ZYLSTRA DAIRY INC
Mailing address: 2655 W. Monte Vista AVE
(not facility address) Modesto CA 95356
Phone#:

Authorized Copy Released To: Copy F+R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/19/23	8:00 AM	W10-1 3831 N DAKOTA	DW	14.4°C		H1				
2	↓	↓	W10-2 3831 N DAKOTA		14.6°		↓				
3			W10-5 4154 Murphy		15.0°C						
4			W10-4E DAKOTA IRRIGATION WELL	IW		14.6°C					H7
5			W9-1E McCARTHY IRRIGATION WELL	IW		15.0°C					H7
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	12-19-23 10:57			T3353008

Required to be submitted to any agency? (YES / NO)

zylstra Dairy 2 2023 Annual Report

Final Audit Report 2024-06-26

Created:	2024-06-26
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAELoHyPJcYMAO7CNSkEJY4ZXzvcV7fOr6

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Document created by Joe Ramos (jramos@fragservices.com)
2024-06-26 - 4:32:23 PM GMT

Document emailed to zylstradairies@gmail.com for signature
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Signer zylstradairies@gmail.com entered name at signing as Mike Zylstra
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